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### RISK ASSESSMENT SYSTEM FOR IDENTIFYING DATA FILES WITH SENSITIVE INFORMATION

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#### Abstract

A system and method are provided for assessing whether data files contain sensitive information associated with an entity. The system stores search keywords associated with the entity, generates search terms based on the search keywords, and searches one or more online public databases for data files associated with each search term. The system then generates risk scores for data files in the search results indicating a likelihood that the data files contain information from a data breach associated with the entity. The system identifies data files that contain information from the data breach from the generated risk scores, and transmits a notification to the entity describing the identified data files.

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/319,030, filed May 17, 2023, which application claims the benefit of U.S. Provisional Patent Application No. 63/343,338, filed May 18, 2022, which is incorporated by reference.

### **BACKGROUND**

[0002] People and organizations may store sensitive information in computing systems. If this information is made available to the public, there could be significant, detrimental impacts to the entity to whom the information belongs. For example, if a person's bank account information were made public, malicious actors could try to take money out of the person's bank account. Similarly, if an organization's customer list were made public, the organization could lose customers to competitors who try to undercut their prices.

[0003] Unfortunately, data breaches do occur on a regular basis, and frequently an entity does not know that a data breach occurs until after they experience negative consequences from their sensitive information being made public. To identify whether a data breach has occurred and what information, if any, has been leaked in a data breach, an entity may want to know whether their information has been made public in publicly available databases. However, it can be very difficult for an entity to check all publicly available databases for whether their information has been published in those places, simply because there is so much additional information for the entity to check. Thus, entities often have to wait until they suffer from the negative consequences of their sensitive information being made public before they know what information is available in the public.

### **SUMMARY**

[0004] A risk assessment system identifies files on online public databases that may contain sensitive information that was leaked from an entity. The risk assessment system stores search keywords for an entity. Search keywords are keywords that describe characteristics of the entity. For example, if the entity is a person, the search keywords may include the person's name, their address, a PIN, bank account numbers, username, password, or personal identifying number (e.g., a Social Security Number). Similarly, if the entity is an organization, the search keywords for the entity may include the name of the organization, the names of members in the organization, bank account numbers associated with the members, or usernames or passwords associated with the organization or its members.

[0005] The risk assessment system generates search terms for an entity based on the search keywords and searches online public databases based on these search terms. A search term for an entity is a combination of search keywords associated with that entity. For example, a search term for a person-entity may include their last name, their address, and their username without including their password or personal identifying number. The risk assessment system uses these search terms to generate search results for the entity based on the data files that the risk assessment system receives through searching the online public databases using the search terms.

[0006] The risk assessment system generates a risk score for each data file in the search results. A risk score indicates a likelihood that the data file includes information that came from a data breach

associated with the entity. The risk assessment system generates the risk scores by applying a risk scoring model to the set of search results. The risk assessment system identifies data files from the search results that contain information from the data breach based on the generated risk scores and transmits a notification to the entity describing the identified files.

[0007] By using search terms generated based on search keywords that are associated with an entity, a risk assessment system can quickly and effectively identify data files that may contain information from an entity's data leak without releasing other private information about the entity. The risk assessment system can also quickly identify data leaks that an entity has experienced before the entity may ordinarily discover the leak, thereby allowing the entity to address issues that may arise from a data leak before the issues actually occur.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Figure (FIG. 1 illustrates an example system environment for a risk assessment system **130**, in accordance with some embodiments.

[0009] FIGS. 2A and 2B illustrate an example data flow where search keywords are used to identify data files that are likely to contain sensitive information, in accordance with some embodiments.

[0010] FIG. 3 is a flowchart for a method of assessing whether data files contain sensitive information for an entity, in accordance with some embodiments.

### DETAILED DESCRIPTION

[0011] Figure (FIG. 1 illustrates an example system environment for a risk assessment system **130**, in accordance with some embodiments. The system environment illustrated in FIG. 1 includes an entity system **100**, one or more public databases **110**, a network **120**, and a risk assessment system **130**. Alternative embodiments may include more, fewer, or different components from those illustrated in FIG. 1, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.

[0012] An entity system **100** is an online computing system that is associated with an entity. An entity may be a person or an organization (e.g., a business, a charity, or a school). The entity system **100** may store data used by the entity. The data stored by the entity system **100** may include sensitive information that the entity does not want made public. For example, for a person-entity, the sensitive information may include a personal identifier (e.g., a Social Security Number), a bank account number, a home address, or health information. For an organization-entity, the sensitive information may include organization financial information, employee information, customer information or patient information. The entity system **100** may be a system that is operated by the entity or by a third-party on behalf of the entity.

[0013] The entity system **100** may suffer from a data breach. A data breach is an incident when data stored by the entity system **100** is copied, taken, or made public without the entity's knowledge or authorization. A data breach may be caused by a malicious hacker who achieves unauthorized access to information stored by the entity system **100** or may be caused unintentionally by the entity or its agent accidentally making information available to the public. Data breaches may include malware placed on the entity system **100**, a device with access to the entity system **100** being lost or stolen or an unintentional or intentional disclosure of information to a third party.

[0014] An online public database **110** is an online system that stores information and makes the information available to the public (e.g., through a website or an API). An online public database **110** may restrict access to users who have an account with the online public database **110** or may

allow information to be openly accessed by anyone who visits a website associated with the online public database **110**. The online public database **110** may associate information with users who provide information to the online public database **110**. An online public database **110** may store information as a data file. A data file is a set of data stored together based on a file format. For example, a data file may be a TXT, DOCX, PDF, XLSX, SQL, CSV, or ZIP file.

[0015] In some embodiments, the online public databases **110** may include search functionality that allows users to search through the information stored by the online public database **110**. For example, an online public database **110** may allow a user to input a search query and the online public database **110** may generate search results for the user based on the search query.

Additionally, an online public database **110** may provide an index by which a user can browse through the information stored by the online public database **110**.

[0016] An online public database **110** may store information from the entity system **100** that was made available to the public due to a data breach. For example, a user of an online public database **110** may be a malicious actor who executed a data breach on the entity system **100**, and may provide data files with sensitive information from the entity system **100** to the online public database **110** to be made available to the public.

[0017] The entity system **100**, online public databases **110**, and the risk assessment system **130** may communicate with each other via the network **120**, which may comprise any combination of local area and wide area networks employing wired or wireless communication links. In some embodiments, the network **120** uses standard communications technologies and protocols. For example, the network **120** includes communication links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, code division multiple access (CDMA), digital subscriber line (DSL), etc. Examples of networking protocols used for communicating via the network **120** include multiprotocol label switching (MPLS), transmission control protocol/Internet protocol (TCP/IP), hypertext transport protocol (HTTP), simple mail transfer protocol (SMTP), and file transfer protocol (FTP). Data exchanged over the network **120** may be represented using any format, such as hypertext markup language (HTML) or extensible markup language (XML). In some embodiments, all or some of the communication links of the network **120** may be encrypted.

[0018] FIG. **1** also illustrates an example system architecture of a risk assessment system **130**, in accordance with some embodiments. The risk assessment system **130** illustrated in FIG. **1** includes a data store **140**, a term generation module **150**, a results generation module **160**, a risk scoring module **170**, and a file identification module **180**. Alternative embodiments may include more, fewer, or different components from those illustrated in FIG. **1**, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.

[0019] The data store **140** stores data used by the risk assessment system **130** to identify data files on online public databases **110** that are likely to contain sensitive information from an entity. For example, the data store **140** may store a copy of sensitive information from the entity system **100** to use as a comparison for identifying data files. Additionally, the data store **140** may store machine-learning models used by the risk assessment system to identify data files.

[0020] The data store **140** stores search keywords for an entity. Search keywords are keywords that describe characteristics of the entity. For example, if the entity is a person, the search keywords may include the person's name, their address, a PIN, bank account numbers, username, password, or personal identifying number (e.g., a Social Security Number). Similarly, if the entity is an organization, the search keywords for the entity may include the name of the organization, the names of members in the organization, bank account numbers associated with the members, or usernames or passwords associated with the organization or its members. Search keywords also may include website domains, mail domains, API endpoints, or IP addresses associated with the

entity.

[0021] The term generation module **150** generates search terms for the entity that the risk assessment system **130** may use to identify data files with sensitive information from the entity system **100**. A search term for an entity is a combination of search keywords associated with that entity. For example, a search term for a person-entity may include their last name, their address, and their username without including their password or personal identifying number. The term generation module **150** may apply term generation rules when generating search terms for an entity. A term generation rule is a constraint on which search keywords may or may not be included in a search term. For example, a term generation rule may require that certain search keywords be included, such as a person's name or username.

[0022] The results generation module **160** generates search results for the entity based on the search terms generated by the term generation module **150**. The results generation module **160** may use the generated search terms to search through online public databases **110** to find data files that match the generated search terms. For example, the results generation module **160** may use a search functionality of an online public database **110** to have the online public database **110** generate a set of search results. Similarly, the results generation module **160** may use the search terms to identify data files on online public databases **110** that may be relevant to the entity. The results generation module **160** may aggregate search results from each of the search terms into an aggregated set of search results for the entity. In some embodiments, the results generation module **160** downloads a set of data files from an online public database **110** and locally searches the downloaded data files. The results generation module **160** may generate search results from the downloaded set of data files based on search terms generated by the term generation module **150**.

[0023] The risk scoring module **170** generates risk scores for each data file in the generated search results. A risk score represents a likelihood that a data file contains sensitive information from an entity's data breach. The risk scoring module **170** may use a risk scoring model to generate a risk score for each data file in the generated search results. A risk scoring model is a machine-learning model (e.g., a neural network) that is trained to generate a risk score for a data file. For example, the risk scoring model may include a transformer model that is trained to generate a risk score for a data file by performing natural language processing on a data file. The risk scoring module **170** may train the risk scoring model based on a set of training examples. Each training example may include an example data file and a label indicating whether the data file contains sensitive information from a data breach. The risk scoring module **170** may iteratively train the risk scoring model by applying the risk scoring model to each data file the training examples and comparing the output of the risk scoring model to the label.

[0024] In some embodiments, the risk scoring module **170** uses a set of machine-learning models to generate risk scores for data files. For example, the risk scoring module **170** may use a file typing model to identify a file type for a data file. A file typing model is a machine-learning model (e.g., a neural network) that is trained to identify a data file type for a data file. File types may describe how the data file is formatted, such as whether the file is a TXT, DOCX, PDF, XLSX, SQL, CSV, or ZIP file. Additionally, file types may describe what kind of information is stored in the file, such as whether the file contains financial information, health information, or employment information.

[0025] The risk scoring module **170** may then use risk scoring models that are trained to generate risk scores for data files of an associated file type. The risk scoring module **170** may use a risk scoring model for each file type. The risk scoring model for a file type may be trained to generate risk scores for data files of the associated file type. For example, the data store **140** may store a risk scoring model that is trained to generate risk scores for PDF files that contain financial information. The risk scoring module **170** may apply a risk scoring model to each data file in the search results based on the file type of each data file and thereby generate risk scores for the data files in the search results.

[0026] The file identification module **180** identifies which, if any, of the data files from the search results contain sensitive information from a data breach. For example, the file identification module **180** may identify data files associated with risk scores that exceed a risk score threshold as data files that likely contain sensitive information from a data breach. Similarly, the file identification module **180** may rank data files by their risk scores to identify data files that likely contain sensitive information.

[0027] The file identification module **180** may transmit the identified files to the entity system **100**, identifying the data files as files that likely contain sensitive information. The file identification module **180** may include information describing the likelihood that the file contains sensitive information, which online public database **110** each data file was retrieved from, or what information is known about whoever provided the data file to the online public database **110**. In some embodiments, the file identification module **180** transmits a ranking of the data files to the entity system **100**.

[0028] FIGS. 2A and 2B illustrate an example data flow where search keywords **200** are used to identify data files that are likely to contain sensitive information, in accordance with some embodiments. FIG. 2A illustrates a set of search terms **210** that are generated based on a set of search keywords **200** that are associated with an entity. In FIG. 2B, those search terms **210** are transmitted to an online public database **110** to be used to generate search results **220**. As illustrated, the search results **220** may include a set of data files from the online public database **110**. The risk assessment system may apply a risk scoring model **230** to the search results **220** to generate risk scores **240** for each data file in the search results **220**. The risk scores **240** may indicate a likelihood that each file contains information from the entity that was part of a data breach. The file identification module **180** identifies data files **250** that are likely to contain information from a data leak based on the risk scores **240**, and may transmit a notification to the entity describing the identified data files **250**.

[0029] FIG. 3 is a flowchart for a method of assessing whether data files contain sensitive information for an entity, in accordance with some embodiments. Alternative embodiments may include more, fewer, or different steps from those illustrated in FIG. 3, and the steps may be performed in a different order from that illustrated in FIG. 3. Additionally, each of these steps may be performed automatically by the risk assessment system without human intervention.

[0030] The risk assessment system stores **300** search keywords for an entity. Each search keyword may describe a characteristic of the entity. The risk assessment system generates **310** search terms based on the search keywords. Each search term may include a different combination of one or more of the search keywords. The risk assessment system may generate the search terms based on term generation rules that apply constraints on which search keywords may or may not be included in a search term.

[0031] The risk assessment system generates **320** search results based on the search terms by searching online public databases based on the search terms. The search results may include data files stored by the online public databases. The risk assessment system generates **330** risk scores for the data files in the search results. A risk score may indicate a likelihood that a data file includes information from a data breach of the entity. The risk assessment system may generate the risk scores by applying a risk scoring model to the data files.

[0032] The risk assessment system identifies **340** data files that contain information from a data breach based on the generated risk scores and transmits **350** a notification to the entity describing the identified data files.

#### ADDITIONAL CONSIDERATIONS

[0033] The foregoing description of the embodiments has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the patent rights to the precise pages disclosed. Many modifications and variations are possible in light of the above disclosure.

[0034] Some portions of this description describe the embodiments in terms of algorithms and

symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

[0035] Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In some embodiments, a software module is implemented with a computer program product comprising one or more computer-readable media containing computer program code or instructions, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described. In some embodiments, a computer-readable medium comprises one or more computer-readable media that, individually or together, comprise instructions that, when executed by one or more processors, cause the one or more processors to perform, individually or together, the steps of the instructions stored on the one or more computer-readable media. Similarly, a processor comprises one or more processors or processing units that, individually or together, perform the steps of instructions stored on a computer-readable medium.

[0036] Embodiments may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, or it may comprise a computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

[0037] Embodiments may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

[0038] The description herein may describe processes and systems that use machine-learning models in the performance of their described functionalities. A “machine-learning model,” as used herein, comprises one or more machine-learning models that perform the described functionality. Machine-learning models may be stored on one or more computer-readable media with a set of weights. These weights are parameters used by the machine-learning model to transform input data received by the model into output data. The weights may be generated through a training process, whereby the machine-learning model is trained based on a set of training examples and labels associated with the training examples. The weights may be stored on one or more computer-readable media, and are used by a system when applying the machine-learning model to new data.

[0039] The language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the patent rights be limited not by this detailed description, but rather by any claims that issue on an application based hereon.

Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the patent rights, which is set forth in the following claims.

[0040] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily

limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive “or” and not to an exclusive “or”. For example, a condition “A or B” is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present). Similarly, a condition “A, B, or C” is satisfied by any combination of A, B, and C having at least one element in the combination that is true (or present). As a not-limiting example, the condition “A, B, or C” is satisfied by A and B are true (or present) and C is false (or not present). Similarly, as another not-limiting example, the condition “A, B, or C” is satisfied by A is true (or present) and B and C are false (or not present).

## Claims

- 1.** A method comprising: storing a plurality of keywords associated with an entity; generating a plurality of search terms based on the plurality of keywords; generating a set of search results by searching one or more data sources based on each of the one or more search terms, wherein the set of search results comprises a set of data files stored by the one or more data sources; generating a risk score for each data file in the set of data files determining whether the data file comprises information that came from a data breach associated with the entity; identifying, based on the generated risk scores for the set of data files, one or more data files of the set of data files that contain information from the data breach; and transmitting a notification to the entity describing the identified one or more data files.
- 2.** The method of claim 1, wherein generating the plurality of search terms comprises: generating a set of different combinations of one or more search keywords from the plurality of search keywords.
- 3.** The method of claim 2, wherein the plurality of search terms are generated based on a set of term generation rules, wherein each term generation rule comprises a constraint on which search keywords of the plurality of search keywords may be included in a search term.
- 4.** The method of claim 1, wherein the one or more data sources comprise one or more online public databases.
- 5.** The method of claim 4, wherein searching the one or more online public databases comprises: transmitting the plurality of search terms to an online public database; and receiving a set of search results from the online public database.
- 6.** The method of claim 4, wherein searching the one or more online public databases comprises searching an indexed set of data files stored by an online public database.
- 7.** The method of claim 1, wherein generating the set of search results comprises aggregating search results from each data source of the one or more data sources.
- 8.** The method of claim 1, wherein generating a risk score for each data file in the set of data files comprises: identifying a file type of each data file in the set of data files; and generating a risk score for each data file in the set of data files by applying a risk scoring model to each data file and to the file type of each data file.
- 9.** The method of claim 8, wherein identifying the file type of each data file comprises applying a file type model to the data file, wherein the file type model is a machine-learning model that is trained to identify a file type of a data file.
- 10.** The method of claim 8, wherein generating a risk score for each data file in the set of data files comprises applying a risk scoring model of a set of risk scoring models to the data file, wherein the risk scoring model is selected from a set of risk scoring models based on the file type of the data file.
- 11.** A non-transitory computer-readable medium storing instructions that, when executed by a computer system, causes the computer system to perform operations comprising: storing a plurality

of keywords associated with an entity; generating a plurality of search terms based on the plurality of keywords; generating a set of search results by searching one or more data sources based on each of the one or more search terms, wherein the set of search results comprises a set of data files stored by the one or more data sources; generating a risk score for each data file in the set of data files determining whether the data file comprises information that came from a data breach associated with the entity; identifying, based on the generated risk scores for the set of data files, one or more data files of the set of data files that contain information from the data breach; and transmitting a notification to the entity describing the identified one or more data files.

**12.** The computer-readable medium of claim 11, wherein generating the plurality of search terms comprises: generating a set of different combinations of one or more search keywords from the plurality of search keywords.

**13.** The computer-readable medium of claim 12, wherein the plurality of search terms are generated based on a set of term generation rules, wherein each term generation rule comprises a constraint on which search keywords of the plurality of search keywords may be included in a search term.

**14.** The computer-readable medium of claim 11, wherein the one or more data sources comprise one or more online public databases.

**15.** The computer-readable medium of claim 14, wherein searching the one or more online public databases comprises: transmitting the plurality of search terms to an online public database; and receiving a set of search results from the online public database.

**16.** The computer-readable medium of claim 14, wherein searching the one or more online public databases comprises searching an indexed set of data files stored by an online public database.

**17.** The computer-readable medium of claim 11, wherein generating the set of search results comprises aggregating search results from each data source of the one or more data sources.

**18.** The computer-readable medium of claim 11, wherein generating a risk score for each data file in the set of data files comprises: identifying a file type of each data file in the set of data files; and generating a risk score for each data file in the set of data files by applying a risk scoring model to each data file and to the file type of each data file.

**19.** The computer-readable medium of claim 18, wherein identifying the file type of each data file comprises applying a file type model to the data file, wherein the file type model is a machine-learning model that is trained to identify a file type of a data file.

**20.** The computer-readable medium of claim 18, wherein generating a risk score for each data file in the set of data files comprises applying a risk scoring model of a set of risk scoring models to the data file, wherein the risk scoring model is selected from a set of risk scoring models based on the file type of the data file.

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